Predicting Future Outcomes

Background / context of the business scenario

Turtle Games, a global toy and games manufacturer, is looking to optimize its sales performance through predictive analysis. One of its core objectives is to understand how their customer demographics and spending behaviours influence loyalty program engagement. This report details an analytical approach using Python and R to develop customer segmentation and construct a Multiple Linear Regression model to predict loyalty point accumulation. By analysing these patterns as well as customer reviews, Turtle Games aims to refine its marketing strategies and implement actionable improvements to its business.

Analytical approach

The analytical workflow was initially executed in Python, utilizing Pandas and NumPy for data manipulation and numerical operations. Matplotlib and Seaborn were employed for data visualization, while Statsmodels and Scikit-learn facilitated the development of Linear Regression, Decision Tree Regressor, and K-Means clustering models. To address the qualitative objectives, NLTK, TextBlob, and WordCloud were used for sentiment analysis and text processing of customer reviews.

Upon import, the dataset was converted into a Data Frame and sense checked to ensure a clean foundation for modelling. Data integrity was verified using the isna().sum() method – confirming zero missing values or NaN values – while info() and describe() were leveraged to validate data types and identify anomalies. Initial cleaning involved dropping the 'language' and 'platform' columns, as their lack of variance offered no analytical value. To reduce potential syntax errors, the "remuneration (k£)"e and "spending_score (1-100)" columns were renamed for brevity. The resulting cleaned dataset was exported as 'turtle_reviews_cleaned.csv' to serve as the single source of truth for the analysis.

Simple Linear Regression models were built first to evaluate how age, remuneration, and spending scores affect loyalty points. The strength of these relationships was gauged based on a combination of R-squared values, p-values, and the extraction of Standard Errors. Visual scatterplots were then used to confirm these statistical trends.

A Decision Tree Regressor was next implemented to assess potential non-linear relationships and evaluate the predictive power of demographic variables, such as gender and education, alongside financial metrics.

Categorical features were first transformed via ordinal mapping to ensure the model could interpret the inherent hierarchy in the data. The data was then partitioned using a 70:30 train-test split prior to model fitting. The initial unpruned model was evaluated using R-squared, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). To improve model robustness and prevent overfitting, pruning was applied by restricting the maximum tree depth to 4.

K-Means clustering was utilized next to segment the base into distinct groups. The final number of clusters was determined using a combination of the Elbow Method and the Silhouette Method.

After which, a comprehensive Natural Language processing pipeline was established to extract actionable insights from customer reviews.

The review and summary columns were isolated and standardized by converting all text to lowercase and removing punctuation. Duplicates were removed to create a corpus for analysis. Tokenization was then applied to split the text into individual words, followed by the removal of NLTK stopwords and alphanumeric characters. A frequency distribution was generated to rank the 15 most common words, providing a high level view of recurring themes. The TextBlob library was leveraged to calculate polarity and subjectivity for each review and summary.

R was subsequently utilized for exploratory data analysis and advanced statistical validation. Histograms, box plots, and scatter plots were used to identify distributions, patterns and outliers. Following feature selection, a Multiple Linear Regression (MLR) model was developed and evaluated using the t-statistics, p-values, Adjusted R-squared and Residual Standard Error (RSE).

The loyalty points distribution was further scrutinized using Q-Q plots, Shapiro-Wilk tests, and skewness/kurtosis functions. Due to the observed right-skew and high kurtosis, a log transformation was performed to stabilize variance and mitigate the influence of extreme values. A final refined MLR model was then constructed and vetted against the same regression metrics.

<u>Visualization and Insights</u>

Seven key visuals were strategically selected to address the core business objectives of identifying loyalty point drivers, segmenting customer personas, and evaluating the predictive reliability of the dataset.

A Feature Importance Bar Plot was initially utilized to establish a hierarchy of influence; it revealed that demographic features (Age, Gender, and Education) held negligible predictive power, providing the rationale to focus exclusively on Remuneration and Spending Score. To quantify these drivers, Linear Regression Scatter Plots demonstrated a strong positive correlation between spending, remuneration, and loyalty point accumulation, while highlighting the variance of errors (the fan shape) that necessitated a log-transformation for model accuracy.

To facilitate targeted marketing strategy, a K-Means Cluster Scatter Plot color-coded the five customer personas. This was paired with a Pie Chart to visualize the proportional distribution of the base, identifying core customer and premium customer groupings.

Finally, a WorldCloud and Polarity Histogram were used to bridge quantitative data with qualitative brand perception. The WordCloud identified recurring word themes, while the Histogram provided a pulse on customer satisfaction. Together, these visuals confirm a stable, neutral to positive sentiment.

<u>Patterns and predictions</u>

The analysis revealed four critical patterns that dictate Turtle Games' sales trajectory. First, Spending Score and Remuneration were identified as the primary engines of the loyalty program, collectively explaining approximately 80% of the variance in loyalty point accumulation (Adjusted R-Squared: ~0.79). We predict that for every 1-point increase in spending score, loyalty points will increase by 2.8%, while every unit increase in remuneration yields a 2.3% increment. This allows the sales team to forecast individual customer lifetime value with high precision ($p < 0.001$).

Secondly, the K-Means Clustering revealed a massive core customer base (Segment 1, 38.7%) with average remuneration and moderate spending. Investigating the price-elasticity of this segment is vital to determine if the customers are motivated by price-points or could be upscaled. Non-linear patterns in Segment 0 versus Segment 2, and Segment 3 versus Segment 4, reveal that high and low remuneration groups both manifest polarized spending habits. This behaviour aligns with observed heteroscedascity (fanning effect) in the linear regression plot: income alone does not predict loyalty points.

Thirdly, as a follow up to the second point, the Residual Standard Error (0.4579) and persistent wave pattern in residuals indicate the model has reached its limit with numerical features and suggests that there is a missing variable. We predict that integrating categorical data is necessary to resolve this systemic variance and achieve higher predictive accuracy.

Finally, the Sentiment Polarity Histogram reveals a distribution concentrated between the Neutral (0.0) and Positive (0.5) ranges. This indicates a stable, low risk brand environment with minimal customer dissatisfaction. However, the high density of neutral scores suggests that a significant portion of the customer is satisfied but not yet highly engaged.